

Ng Tze Kean (吴子健)

| ✉ ngtzekean@gmail.com | ☎ (+65) 8228 8710 | 🌐 hiiamtzekean | 🌐 ngtzekean |

🏢 PROFESSIONAL EXPERIENCE

Senior Data Scientist

Flo Energy Pte Ltd | Supervisor: Nils Trepstra [Head of Data & Analytics]

Apr 2026 – Present

Singapore

Data Scientist

Flo Energy Pte Ltd | Supervisor: Nils Trepstra [Head of Data & Analytics]

Jun 2025 – Mar 2026

Singapore

- Handling of noisy, non-stationary and volatile nature of data to model volatility of energy system
- Approximating data backfill from distribution profile
- Energy load forecasting via statistical methods to construct most probable consumption pattern in various states of Australia. Some methods include quartile forecasting, long/short term time series modelling
- Time series handling of PV solar data. Decomposition for detrending and effect removal to support forward forecast of PV load
- Effect removal and smoothing algorithms to distribute load spikes for materialisation of negative/positive impact to PnL in a controlled manner
- Robust optimization models for uncertain market prices and load, using defined uncertainty sets to support business decisions that reduce downside risk under adverse market conditions
- Building PnL models for downstream users to formulate position to hedge against

Data Analyst

Flo Energy Pte Ltd | Supervisor: Nils Trepstra [Head of Data & Analytics]

Dec 2024 – Jun 2025

Singapore

- Predictive modelling of demand and supply of the SG energy market through bid ask simulation for spot price and day ahead forecast
- Battery Energy Storage Systems (BESS) simulation and analysis of arbitrage opportunity to drive potential new revenue stream
- Feature engineering and time series analysis in Python to build ML such as XGBoost and tree based models. Further exploration was also done to test feasibility of DNN models like LSTM and Attention networks to capture temporal and recurrence nature of data
- Data engineering work using Dagster for ETL purposes to transform data pulled from API into data lake which is later then transformed into data cube
- Orchestrated robust data flows from data lake to Snowflake warehouse, including schema design, incremental loading strategies for downstream usage of further modelling of energy demand and price
- Python concurrent and parallel programming to optimise program compute time

Data Analyst Intern, ADDX Pte Ltd

Singapore, Jun 2023 – Dec 2023

Data analytics, dashboards, predictive models

Finance Intern, Wiz.AI Pte Ltd

Singapore, May 2023 – Jun 2023

Financial tracking and account statements

Data Analyst Intern, REANGLE Pte Ltd

Singapore, May 2022 – Aug 2022

Green finance analytics and ESG digital marketing

Engagement & Strategy Intern, MOH

Singapore, Nov 2020 – Apr 2021

Telemedicine E-training, process automation

Platoon Commander, Singapore Armed Forces

Singapore, Jan 2019 – Nov 2020

Top Graduate, Sword of Honour

🎓 EDUCATION

Nanyang Technological University

Bachelor in Computer Science

Aug 2021 – Jul 2025

Singapore

- CGPA: 4.93/5.0 (Highest Distinction Honours) | Dean's List all years applicable
- Specialization in AI, Minor in Business
- Valedictorian of class 2025, Koh Boon Hwee Scholar Awardee 2025

🌐 EXCHANGE UNIVERSITIES

Peking University

China, Jul 2024 – Jul 2024

Scholarship offered for summer program

Tsinghua University

Singapore (Virtual), Aug 2023 – Dec 2023

Big data systems and deep learning

Shanghai Jiao Tong University

China, Jan 2024 – Jun 2024

Reinforcement learning and Cyber security

National University of Singapore

Singapore, Jan 2023 – May 2023

Software engineering and Database systems

University Michigan - Shanghai Jiao Tong University

China, Jan 2024 – Jun 2024

Machine learning in Python

🔗 RESEARCH PROJECTS

RecNextEval: A Reference Implementation for Temporal Next-Batch Recommendation Evaluation Dec 2025 – Apr 2026

Published paper in SIGIR 2026 | Supervisor: Prof. Sun Aixin [Associate Dean Academics]

- Published at arXiv:2604.13665v1

Streamsight: A toolkit for offline evaluation of RS

Jan 2024 – Dec 2024

FYP, grade obtained: A+ | Supervisor: Prof. Sun Aixin [Associate Dean Academics]

- Python, Recommender System, System architecture design, Machine learning techniques
- Evaluation framework to account for temporal dynamics of user/item interactions through sliding window implementation and streaming of dataset to RecSys to capture realistic data communication

Improving Localization Robustness of ORB-SLAM with Dual Cameras

Aug 2022 – Jun 2023

NTU President Scholar (Merit) | Supervisor: Prof. Lam Siew Kei [Assistant Chair Admission]

- Embedded system (Nvidia Jetson Nano), GStreamer, ROS, ORB-SLAM, OpenCV, Linux
- Develop dual-camera framework to be used in conjunction with ORB-SLAM to increase the feature points for better localization of agent

🏠 SCHOOL PROJECTS

AI Related

- Pytorch, Tensorflow implementation of recurrent and transformer models for sentiment analysis
- LLMs from huggingface for data augmentation to support downstream tasks

Data Science Related

- EDA and machine learning (predictive modeling) to forecast future prices
- B+ tree, Query planning implementations in database management systems

🏆 AWARDS

- Valedictorian of college (2025)
- Koh Boon Hwee Scholar Award (2025)
- Genk Capital Scholarship (2024)
- Tan Sri Runme Shaw Scholarship (2023)
- Ng Bok Eng Scholarship (2023)
- Dean's list all years applicable

🏆 HACKATHONS

- China Global Innovation Competition Joint prosthesis (2024)
- NTU SCSE TechFest Hackathon, 1st runner up (2024)
- TikTok Hackathon Challenge, Finalist (2023)
- SUTD Hackathon, Champion (2023)

💼 BUSINESS MODULES

- Investments
- Business Finance
- Accounting
- Business Law
- Management

💻 COMPUTING MODULES

- Reinforcement learning
- Big data management, Database
- Natural language processing
- Machine learning, Deep learning
- Operating system

🌟 QUALITIES

- MBTI: ENTJ
- English, Chinese proficient
- Worked in cross functional teams
- Python, SQL, Java, C
- Env (Window, MacOS, Linux)

👤 LEADERSHIP

- NTU Ambassador 2024 - 2025
- NTU College (CCDS) Ambassador 2021-2024
- NTU Overseas (GEM) Ambassador 2024
- NTU College (COE) Ambassador 2021-2023
- EXCO, NTU-Dyson ambassador 2021-2022
- EXCO, NTU Aikido 2021-2022

★ MAJOR ACHIEVEMENTS

- Appointed and represented NTU locally and globally in key partnership events such as Singapore-Sweden dialogue in NTU where I hosted the King of Sweden.
- Gathered over 48 donors and raised 1.49k for NTU Bursary fund raising in 2023 and subsequently invited to join the panelist in NTU giving day to share more about the importance of giving back to the community.
- Dean's list all years applicable along with multiple scholarships and awards from NTU and partner organisations.
- Invited to consistently participate in NTU's leadership and ambassadorial programs to represent the student body in fund raising and donor appreciation.

💧 CURRENT INTERESTS

- Design and building of predictive model architecture for forecasting and enhancing decision making
- Applications in NLP through langchain to create agentic workflows